

File Structure and File Guide

Every project file, what it does, and which file to open when a teacher asks.

16 September 2026 | BhoomiScanV18

1. How to use this guide in viva

When the teacher says 'show me X', use Section 2 first. Then open the file path in Cursor (Ctrl+P and type the file name). Do not open node_modules, .venv, or frontend/dist. Those are libraries and build output, not your code.

2. Teacher asks... open this file

Login / JWT / signup	backend/app/routers/auth.py and frontend/src/pages/Login.jsx
Database tables	backend/app/models/ (user, document, analysis, report)
Upload PDF	backend/app/services/storage.py and frontend/src/pages/DocumentUpload.jsx
Sarvam AI / OCR	backend/app/services/sarvam_service.py
Analyze pipeline	backend/app/services/analysis_service.py
Litigation / risk	backend/app/services/risk_engine.py
Seller / buyer names	backend/app/services/extraction_service.py
Risk score 70	backend/app/services/scoring_service.py and constants.py
Blockchain / hash	backend/app/services/hashing.py and ledger_service.py
Show hash as a file	storage/ledger/CHAIN.json
VALID / TAMPERED	backend/app/services/report_service.py
Simple report screen	frontend/src/pages/FinalReport.jsx
API routes list	backend/app/main.py and frontend/src/App.jsx
Phase 1 tests	backend/tests/test_phase1.py
Phase 2 tests	backend/tests/test_phase2.py
Phase 3 tests	backend/tests/test_phase3.py
Navy / gold theme	frontend/src/index.css and tailwind.config.js
Env / API key	root .env (never commit) and .env.example
Postgres Docker	docker-compose.yml

3. Folder tree (your code only)

BhoomiScanV18/

README.md, .env, .env.example, docker-compose.yml, .gitignore

backend/ FastAPI API (port 8000)

requirements.txt, .venv/

app/ application code

main.py, config.py, constants.py, database.py, deps.py

models/ SQLAlchemy tables

schemas/	request/response shapes
routers/	HTTP endpoints
services/	business logic
tests/	pytest Phase 1, 2, 3
frontend/	React + Vite (port 5173)
index.html, package.json, vite.config.js, tailwind.config.js	
src/pages/	screens
src/components/	reusable UI
src/hooks/, services/, constants/	
storage/documents/	uploaded PDFs on disk
storage/ledger/	CHAIN.json hash-chain files
docs/	test reports and this guide

Not listed below: backend/.venv, node_modules, frontend/dist, __pycache__. Those are generated.

4. Root files

README.md

How to start Postgres, backend, and frontend.

.env

Secrets on your PC: DATABASE_URL, JWT_SECRET, SARVAM_API_KEY. Do not put this in Git.

.env.example

Safe template of env names without real keys. Show this, not .env.

docker-compose.yml

Starts PostgreSQL 16 as container bhoomiscan-db on port 5432.

.gitignore

Keeps .env, venv, node_modules, and uploaded files out of Git.

5. Backend core

backend/requirements.txt

Python packages: FastAPI, SQLAlchemy, pytest, sarvamai, fpdf2, bcrypt.

backend/app/main.py

App entry. Creates tables, mounts routers, health check GET /health.

backend/app/config.py

Reads .env. Database URL, JWT, upload size, storage folder, Sarvam wait times.

backend/app/constants.py

Enums: document types, risk statuses DETECTED/NOT_VERIFIED/NO_ISSUE_FOUND, score HIGH=70, GENESIS, disclaimer.

backend/app/database.py

SQLAlchemy engine and get_db() session for Postgres (tests use SQLite).

backend/app/deps.py

get_current_user: reads Bearer JWT. Protects all case/report routes.

backend/app/db_migrate.py

Adds new columns on startup so old Phase 1 rows keep working.

backend/app/__init__.py

Marks app as a Python package.

6. Backend models (database tables)

Teacher: 'Where is data stored?' Open models, then say PostgreSQL database bhoomiscan.

backend/app/models/__init__.py

Imports all models so tables are created together.

backend/app/models/user.py

Table users: username, bcrypt password hash.

backend/app/models/property_case.py

Table property_cases: one file per property (domain, type, owner).

backend/app/models/document.py

Table documents: filename, disk path, OCR text, extracted JSON, type.

backend/app/models/analysis.py

Tables analyses, risk_findings, evidence, recommendations.

backend/app/models/report.py

Tables document_comparisons, final_reports (JSON + SHA-256), ledger_records (previous_hash, record_hash).

7. Backend schemas (API JSON shape)

backend/app/schemas/auth.py

Signup/login request and token response.

backend/app/schemas/property_case.py

Create-case body: domain and property type.

backend/app/schemas/document.py

Upload metadata returned to the UI.

backend/app/schemas/analysis.py

Analysis, findings, evidence JSON for the analysis page.

backend/app/schemas/report.py

Recalculate payload, final report, integrity VALID/TAMPERED.

backend/app/schemas/__init__.py

Package marker.

8. Backend routers (URLs)

backend/app/routers/__init__.py

Package marker.

backend/app/routers/auth.py

POST /auth/signup, /auth/login, GET /auth/me.

backend/app/routers/property_cases.py

Create case, list cases, upload file, list/delete documents.

backend/app/routers/analysis.py

POST analyze, GET analysis, recommendations, evidence.

backend/app/routers/reports.py

Recalculate, due-diligence, finalize, list reports, verify, PDF download.

9. Backend services (logic)

Teacher: 'Where is the AI / risk / blockchain?' This folder. Routers call these; they do not talk to the UI directly.

backend/app/services/auth.py

Hash password with bcrypt. Create and decode JWT.

backend/app/services/storage.py

Save uploaded PDF/PNG/JPG under storage/documents/<case-id>/.

backend/app/services/sarvam_service.py

Live Sarvam: digitise (OCR), translate Kannada to English, extract JSON. Tests mock this.

backend/app/services/document_classifier.py

Guesses document type from keywords (sale deed vs court paper).

backend/app/services/extraction_service.py

Reads survey, seller, buyer, case number from text. Fixes Indian deed 'executed by' names.

backend/app/services/evidence_service.py

Cuts source snippets around extracted values. Never hides missing names.

backend/app/services/risk_engine.py

Five checks: ownership, litigation, mortgage, approval, property record. DETECTED / NOT_VERIFIED / NO_ISSUE_FOUND.

backend/app/services/recommendation_engine.py

Suggests missing papers (EC, parent deed, plan approval).

backend/app/services/analysis_service.py

One-document pipeline: OCR -> translate -> classify -> extract -> risk -> save.

backend/app/services/comparison_service.py

Compares fields across documents: MATCH / MISMATCH / NOT_AVAILABLE.

backend/app/services/scoring_service.py

Risk score. DETECTED HIGH = 70 (litigation). Reasons list. Coverage math. NOT_VERIFIED adds 0.

backend/app/services/case_analysis_service.py

Recalculate whole case from all analyzed papers. Merges findings.

backend/app/services/hashing.py

Canonical JSON then SHA-256. Same content always same hash.

backend/app/services/ledger_service.py

Local hash chain. First previous_hash = GENESIS. Writes CHAIN.json too.

backend/app/services/report_service.py

Finalize: store report JSON, hash it, append ledger, verify VALID or TAMPERED.

backend/app/services/report_pdf.py

Builds the downloadable risk-report PDF (score, because, hash).

backend/app/services/__init__.py

Package marker.

10. Backend tests

backend/tests/test_phase1.py

13 tests: signup, login, JWT, residential case, upload, 401/403. No AI.

backend/tests/test_phase2.py

7 tests: analyze auth, mocked Sarvam, litigation DETECTED, missing evidence is NOT_VERIFIED.

backend/tests/test_phase3.py

19 tests: vault, MATCH/MISMATCH, score 70, GENESIS ledger, VALID, TAMPERED, names extraction.

11. Frontend config

frontend/package.json

npm scripts: dev, build. React 19, Vite 6, Tailwind, Router, Axios, Framer Motion.

frontend/package-lock.json

Exact library versions. Do not edit by hand.

frontend/vite.config.js

Vite bundler. Dev server <http://localhost:5173>.

frontend/tailwind.config.js

Colors: royal navy, gold, ivory, void.

frontend/postcss.config.js

Required for Tailwind.

frontend/index.html

HTML shell. Title: BhoomiScan.

frontend/.env.example

VITE_API_URL=<http://localhost:8000>

frontend/.env

Local API URL for the browser. Only VITE_ names are exposed.

frontend/src/main.jsx

React start: Router, AuthProvider, ToastProvider.

frontend/src/index.css

Global dark theme, royal-frame cards, gold line.

frontend/src/App.jsx

All routes. Protected pages need login.

12. Frontend pages (screens)

frontend/src/pages/Landing.jsx

Public home. Your Alvocate. Get Started / Login.

frontend/src/pages/Signup.jsx

Create username + password.

frontend/src/pages/Login.jsx

JWT login. Demo Google/X/phone buttons are not real OAuth.

frontend/src/pages/Home.jsx

Dashboard after login. Start analysis.

frontend/src/pages/Dashboard.jsx

Older/alias dashboard pieces if used.

frontend/src/pages/PropertyDomain.jsx

Residential vs Agricultural/Commercial/Industrial. Last three go to Get Subscription.

frontend/src/pages/PropertyType.jsx

Villa, plot, independent house, etc.

frontend/src/pages/DocumentUpload.jsx

Document vault. Upload, analyze, delete, Get risk report.

frontend/src/pages/DocumentAnalysis.jsx

One-document AI result. Risk cards. Document text toggle (hidden by default).

frontend/src/pages/DueDiligence.jsx

Does not show the old complex page. Recalculates, finalizes, jumps to the simple report.

frontend/src/pages/FinalReport.jsx

THE report: Risk score, Because reasons, VALID integrity, Download PDF.

frontend/src/pages/AI.jsx

In-app chat page (demo helper, not Sarvam analysis).

frontend/src/pages/Profile.jsx

User profile.

frontend/src/pages/ProfileSettings.jsx

Profile settings.

frontend/src/pages/Contact.jsx

Contact Us.

frontend/src/pages/Subscription.jsx

Get Subscription screen for non-residential domains.

13. Frontend components, hooks, services

frontend/src/components/AppHeader.jsx

Top bar: logo, Home, AI, profile menu. Sign Out is only inside the menu.

frontend/src/components/Navbar.jsx

Re-exports AppHeader.

frontend/src/components/Logo.jsx

BhoomiScan globe + Your Alvocate.

frontend/src/components/PageShell.jsx

Page layout with header and atmosphere.

frontend/src/components/Atmosphere.jsx

Background glow.

frontend/src/components/Button.jsx

Gold / ghost / primary buttons.

frontend/src/components/Input.jsx

Form fields.

frontend/src/components/Card.jsx

Generic card.

frontend/src/components/ErrorMessage.jsx

Red error text.

frontend/src/components>LoadingSpinner.jsx

Analyzing / signing-in spinner.

frontend/src/components/ProtectedRoute.jsx

Redirects to /login if no token.

frontend/src/components/ProfilePill.jsx

Name chip that opens the menu.

frontend/src/components/ProfileDropdown.jsx

Profile, Settings, Contact, Subscription, Sign Out.

frontend/src/components/UploadBox.jsx

Drag-and-drop file box.

frontend/src/components/DocumentCard.jsx

One uploaded paper with Analyze / delete.

frontend/src/components/FlowStepper.jsx

Steps: property kind, home type, vault, risk report.

frontend/src/components/AnalysisProgress.jsx

Reading / extracting / checking steps.

frontend/src/components/RiskCategoryCard.jsx

One of five risk tiles on analysis.

frontend/src/components/ExtractedDetails.jsx

Survey, seller, buyer after analysis.

frontend/src/components/EvidencePanel.jsx

Source snippets popup.

frontend/src/components/RecommendationList.jsx

Upload EC / parent deed suggestions.

frontend/src/components/ComparisonTable.jsx

MATCH/MISMATCH table (API still has it; simple report UI hides it).

frontend/src/components/ConfirmDialog.jsx

Yes/cancel dialog.

frontend/src/components/PropertyCard.jsx

Domain card on dashboard flow.

frontend/src/components/PropertyTypeCard.jsx

Home-type card.

frontend/src/components/PropertySummaryCard.jsx

Dashboard property summary.

frontend/src/components/ActivityCard.jsx

Recent activity tile.

frontend/src/components/SubscriptionCard.jsx

Plan card.

frontend/src/components/UpgradeCard.jsx

Upgrade prompt.

frontend/src/components/AIChat.jsx

Chat UI.

frontend/src/components/DemoAuthButton.jsx

Fake Google/X/phone control. Not real OAuth.

frontend/src/components/icons.jsx

Small SVG icons.

frontend/src/hooks/useAuth.jsx

Login state. Token in localStorage key bhoomiscan.token.

frontend/src/hooks/useToast.jsx

Popup messages.

frontend/src/hooks/useSubscription.js

Subscription flag for gated domains.

frontend/src/services/api.js

Axios calls to FastAPI including analyze, finalize, verify, PDF.

frontend/src/services/auth.js

Auth helper wrappers.

frontend/src/services/ai.js

Chat helper.

frontend/src/constants/property.js

Domains, property types, token key.

frontend/src/constants/analysis.js

Risk labels for UI.

frontend/src/constants/subscription.js

Subscription copy.

frontend/src/lib/userDisplay.js

Display name initials.

14. Storage, ledger files, docs

storage/documents/<case-id>/

Original uploaded PDFs. Show this for 'where are the papers stored?'

storage/ledger/CHAIN.json

Human-readable hash chain. previous_hash GENESIS then linked hashes. SHOW THIS for blockchain.

storage/ledger/report_v1.json

One finalized report's hash, score, timestamp.

docs/generate_test_reports.py

Builds Phase 1/2/3 test-case PDFs.

docs/generate_project_summary.py

Builds the complete project summary PDF.

docs/generate_file_structure.py

This guide's generator.

docs/Phase1_Test_Case_Report.pdf

Phase 1 test report.

docs/Phase2_Test_Case_Report.pdf

Phase 2 test report.

docs/Phase3_Test_Case_Report.pdf

Phase 3 test report.

docs/BhoomiScan_Complete_Project_Summary.pdf

Long written summary for report/PPT.

15. What to say in one sentence

React UI talks to FastAPI. FastAPI stores users, papers, findings, and reports in PostgreSQL. Sarvam only runs on Analyze. Python decides risk. Finalize SHA-256 hashes the report and appends a local GENESIS ledger, also written to storage/ledger/CHAIN.json. The student UI is a simple score-and-reason report with VALID/TAMPERED integrity.